INCH-POUND

MIL-P-26539D
AMENDMENT 2
17 May 1995
SUPERSEDING
AMENDMENT 1
12 March 1991

MILITARY SPECIFICATION

PROPELLANTS, NITROGEN TETROXIDE

This amendment forms a part of MIL-P-26539D, dated 28 August 1989, and is approved for use by all Departments and Agencies of the Department of Defense

PAGE 1

1.1: Delete and substitute:

"1 1 Purpose This specification covers the requirements for five types and two grades of nitrogen tetroxide (N_2O_4) propellants."

1.2: Delete and substitute:

"1.2 Classification. Nitrogen tetroxide propellants shall be of the following designated types and grades as specified (6.2).

Types

NTO - Nominal 99.5% N_2O_4 purity with red-brown color.
MON-1 - Nominal 99% N_2O_4 and 1% NO with green color.
MON-3 - Nominal 97% N_2O_4 and 3% NO with green color.
MON-10 - Nominal 90% N_2O_4 and 10% NO with green color.
MON-25 - Nominal 75% N_2O_4 and 25% NO with green color.

Grades

Standard - no iron requirement.
Low-iron - maximum 0.5 parts per million iron."

PAGE 2

2.2: Add the following:
"International Civil Air Organization (ICAO)
Technical Instructions for the Safe Transport of Dangerous Goods by Air
International Air Transport Association (IATA)
Dangerous Goods Regulations
International Maritime Organization (IMO)
International Maritime Dangerous Goods Code (IMDG Code)"

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TABLE I: Delete and substitute:

TABLE I. Chemical composition and physical properties.

Composition	NTO (Red-Brown)	MON-1 (Green)	MON-3 (Green)	MON-10 (Green)	MON-25 (Green)	Test Para.
N_2O_4 assay % wt, MIN	99.5	---	---	---	---	4.5.3
NO content % wt, MAX		1 0	3.0	11.0	26.0	4.5 2
% wt, MIN	1/	0.6	2.5	10.0	25.0	
N_2O_4 + NO % wt, MIN	---	99.5	99.5	99.5	99.5	4.5.3
Water equivalent % wt, MAX	0.17	0.17	0.17	0 17	0.17	4.5 4
Chloride content % wt, MAX	0 040 2/	0 040	0 040	0.040	0.040	4.5.5
Nonvolatile residue 3/ mg/L, MAX	---	10.0	10.0	10.0	10.0	4.5.6
Iron content 3/ ppm wt, MAX	---	0.5	0 5	0.5	0.5	4.5.7
Particulate mg/L, MAX	10.0	10.0	10.0	10.0	10.0	4.5.8

1/ The NO content shall be limited to that which does not change the specified red-brown color of the propellant (3.4).

2/ This test need not be performed on NTO propellant if the material was manufactured by the ammonia-oxidation process.

3/ This requirement applies to the low-iron grade of the propellant only "

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4.4.2.1.3: Add the following sentence:

"Each cylinder of low-iron grade material shall constitute a separate lot and shall be sampled."

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4.5.2.1.3: Delete the calculation for NO and substitute:

$$NO, percent\ by\ wt = \frac{\left[y - 0.003857\left(z - \frac{x+y}{149}\right)\right]1875}{x} - 015$$

PAGE 15

4.5.3.2, line 15: Delete "methyl red or methyl orange" and substitute "bromothymol blue".

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4.5.3.4a(1), line 3: Delete "methyl red or methyl orange" and substitute "bromothymol blue".

4.5.3.4a(3): Delete and substitute:

"Bromothymol blue indicator solution: 0.1 grams in 16 mL of 0.01 N NaOH and 234 mL of distilled water."

4.5.3.4a(5), line 2: Delete "methyl red" and substitute "bromothymol blue".

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* 4.5.4.2.1: Delete the third sentence and substitute:

"Set the temperature of a circulating low temperature bath to -6 ± 1°C (21.2 ± 1.8°F).

4.5.4.2 1: Add the following sentence:

"Throughout the procedure, minimize the exposure of the oxidized N_2O_4 to the atmosphere."

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* 4.5.4.2.4b. Delete and substitute:

"Circulating low temperature bath."

4.5.4.2.4f, line 1: Add ", or equivalent." to the end of the line.

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4.5.5.1.1, line 13: Delete "a pH of 2.5 to 3.0 is reached" and substitute "the pH stabilizes between 2.5 to 3.0".

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* 4.5.6.1: Delete and substitute:

"4.5.6.1 Procedure. Thoroughly clean two platinum evaporating dishes per sample. Dry the dishes in an oven at 105 ± 5°C (220 ± 10°F). After

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drying, cover the dishes with a watch glass and allow them to cool next to the balance for 30 minutes. Weigh each dish to the nearest 0.01 milligram Place a hot plate in an exhaust hood and place the weighed dishes on it. Transfer 50 mL of the N_2O_4 to each dish. Using the hot plate, evaporate the sample to near dryness Oven dry the dishes for one hour. Cover the dishes with a watch glass and allow to cool next to the balance for 30 minutes. Weigh each dish to the nearest 0 01 mg Repeat oven drying until constant weight is attained."

* 4 5 6 2: Delete and substitute:

"4 5.6.2 Calculation. The mg/L nonvolatile residue shall be calculated by the following formula:

$$NVR(mg/L) = \frac{W_S - W_E}{V_S}$$

where: W_S = Weight of sample dish after drying (mg)
W_E = Weight of empty dish (mg)
V_S = Volume of sample used (litres)"

PAGE 27

4 5 7, line 1 Delete "MON-1 and the MON-3" and substitute "propellant".

4.5 7.1, line 1: Delete "Using a 100 mL burette".

4.5 7.1, line 2: Delete "MON-1 or MON-3" and substitute "propellant".

4.5 7.2, line 1: Delete "MON-1 or MON-3" and substitute "propellant".

PAGE 29

* 4.5.8 1: Delete and substitute:

"4 5.8.1 Preparation of the membrane filter. Using forceps place a 47-mm membrane filter of nominal pore size (10 µm) in a clean petri dish. Place the petri dish, with its lid slightly ajar, in an oven at 90 ± 5°C (194 ± 9°F) and leave it for 30 minutes. Remove the petri dish from the oven and place it near the balance. The petri dish cover should be ajar but still protecting the membrane filter from contamination from the atmosphere Allow 30 minutes for the membrane filter to come to equilibrium with the ambient air temperature and humidity. Remove the membrane filter from the petri dish with forceps, handling it by the edge only, and place it on the center of the weighing pan. Weigh and return it to the petri dish.

* 4.5.8.2: Delete.

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* 4.5.8 4: Delete and substitute:

"4.5.8.4 Procedure Assemble the high pressure filter holder with the weighed membrane filter and position it so that the filter membrane is in a horizontal position. Pass 400 to 500 mL of sample through the filter holder Follow with clean, dry nitrogen gas to purge remaining N_2O_4. Carefully remove

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the membrane filter, and place it in a clean, covered petri dish. Dry and reweigh the membrane filter taking care not to disturb the contaminant on the its surface. The drying procedure described in 4.5.8.1 should be followed in detail."

* 4.5.8.5: Delete and substitute:

"4.5.3.5 Calculation. Subtract the initial weight of the test membrane filter, W1, from the final weight, W2. Divide the resulting weight by the volume of sample filtered and report the result as total contaminant, expressed in milligrams per litre to the nearest 0.01 mg/L. Also report the volume of sample used in the test.

* 4.5.8.6: Delete and substitute:

"4.5.8.6 Equipment. The following equipment shall apply as test conditions of 4.5.8:

a. Analytical balance, single or double pan, the precision standard deviation of which must be 0.07 mg or better.

b. Oven, of the static type (without fan-assisted air circulation), controlling to 90 ± 5°C (194 ± 9°F).

c. Petri dishes, approximately 50 mm in diameter with removable glass supports for membrane filters.

d. Forceps, flat-bladed with unserrated, non-pointed tips.

e. Membrane filters, 47-mm diameter, polytetrafluoroethylene, filter disc with rating of 10 ± 3 µm.

f. High pressure filter holder, 47 mm.

g. Clean, dry nitrogen, water vapor - 26 ppm by vol max.

h. Air ionizer for the balance case."

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* 5.1: Delete and substitute:

"5.1 Packaging. The product furnished under this specification is a hazardous material. Packaging shall comply with Code of Federal Regulation 49 (49 CFR), DOT Special Permit/Exemption, and United Nations (UN)/International requirements as stipulated in ICAO, IATA, and IMO, as applicable for mode of transportation and destination. The product shall be packaged in containers and unit quantities as specified by the procuring activity (6.2 and 6.3)."

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5.4: Delete and substitute:

"5.4 Labeling and marking. Each container shall be labeled, marked, and placarded in accordance with 49 CFR, DOT Special Permit/Exemption, MIL-STD-

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129, and UN/International requirements as stipulated in ICAO, IATA, and IMO, as applicable for mode of transportation and destination In addition, each container shall be marked "INHALATION HAZARD". An identification tag, precautionary label, and container color code with name identification shall be used on each cylinder

6.2 1, line: Delete "composition" and substitute "composition, grade".

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6.3e, line 1: Delete "MON-1 and MON-3 product" and substitute "propellant".

NOTE The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue

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Civil Agency Interest
NASA

(Project 9135-0128)

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